

What is really driving VoltRelay's service failures, rider churn and eroding margins?

Team WindRocker · 18 months of network data (Jan 2024 – Jun 2025) · 3.81M swap attempts · 152 stations · 6 cities

VoltRelay does not have a demand problem or a "too few stations" problem. It has **three fixable operational problems**: **old Gen1 chargers that overheat** in the Jaipur, Delhi NCR and Hyderabad summers; **three bad battery lots from one supplier (Kyron KY-2407/08/09) that degraded 2.4× faster**, which wiped out the price-rise margin gain; and **new riders who hit repeated failed swaps in their first fortnight** and leave. Fix the equipment, replace the bad packs, protect new riders, and do not lock in the ZipDrop discount.

COMPLETED SWAPS / MONTH

104K → 303K
2.9× in 18 months

REVENUE / MONTH

₹6.2M → ₹19.7M
₹229M over the period

SUMMER FAILURE RATE

8.4%
vs 3.9% rest of year · peak 11.9%

CM / SWAP, H1 2025

-₹17.5
-₹4.2 without the three bad lots

① Heat × Gen1 chargers

Above 45°C a Gen1 cabinet needs **156 min** (not 88) to charge a pack and runs out of stock in **37%** of hours. The **36 hot-city Gen1 stations** handle 32% of summer attempts but cause **57.5%** of summer failures.

② Three bad battery lots

Kyron lots KY-2407/08/09 deliver **49 km vs 61 km** and cost **₹82 vs ₹35** in wear per swap; they explain **~76%** of the H1-2025 margin loss. Kyron's later lots wear normally. Battery/range is now the #1 complaint.

③ First-fortnight failures

New riders with **2+ failed swaps in their first 14 days** churn at **15.0% vs 9.0%** (adjusted OR 1.58, **2.05** with early activity controlled). Price plan, channel and partner do not matter.

How to read this report. Section 1 frames the problem, Section 2 shows how we cleaned and analysed the data, Section 3 answers the six core questions with evidence, Section 4 separates primary from secondary drivers and answers the budget question, and Section 5 lists prioritised, sized actions. Appendix B shows how we tried to break our own findings (confounding, exposure bias, double counting) and what changed as a result. Every chart is reproducible from the two notebooks, and the interactive dashboard lets leadership filter by city, vehicle, charger generation and date.

1 Problem Understanding

Business context. VoltRelay runs unmanned battery-swap cabinets for gig and logistics riders (food delivery, quick commerce, e-commerce, cargo, bike taxi) across six Indian cities. Its riders are paid per trip, so **every minute at a station is lost income**. A failed swap costs VoltRelay a sale and gives the rider a reason to switch networks.

The paradox. Over 18 months completed swaps and revenue grew strongly, but service failures rose faster, new-rider retention fell and per-swap profitability eroded. In the same window VoltRelay expanded in two waves, raised its base price, piloted peak/off-peak pricing in two cities, onboarded a new battery supplier and renegotiated its largest fleet contract. Leadership must choose between four budget proposals (**more stations, more batteries, network-wide peak pricing, a long-term ZipDrop exclusive**) and needs to know what is actually driving the outcomes.

How we framed it: one causal chain, not six separate questions

Equipment & assets	Operations	Service experience	Retention	Unit economics
charger generation, battery supplier/lot, climate	charge time, charged stock, outages	failed / abandoned swaps, range delivered	does a new rider come back?	revenue, energy, battery wear, site cost

For every factor we asked three things: **is there a mechanism** (not just a correlation), **how big is the effect**, and **how many riders or swaps does it touch**. Factors that are large, widespread and consistent across cuts are called **primary drivers**; real but smaller or narrower effects are **secondary/contributing**. The data is observational, so we argue *likely* causes from mechanism, timing and dose-response, and state where that stops.

Core question	What we measured	Headline answer
Q1 Network performance over time	Monthly completed swaps, revenue, failure rate, fully-loaded contribution margin (CM) per swap	Volume ↑ 2.9×; failures spike every summer; CM gain from the price rise erased by battery wear
Q2 Service failures & experience	Failure mix by hour, season, city, vehicle class; station concentration; link to hourly stock	Concentrated: hot cities, evening rush, Gen1 sites, 3W
Q3 Stations & geography	Charger generation, cabinet temperature, charge time, stock-outs, tickets per 1,000 swaps	Gen1 thermal derating above 45°C is the mechanism
Q4 Batteries & equipment	SoH trajectories by supplier/lot/BMS, delivered km, wear cost per swap	Kyron lots KY-2407/08/09 are the outlier cohort
Q5 Pricing & partners	Base-price step, peak-pilot difference-in-differences, partner CM per swap	Pricing = revenue lever, not congestion lever; ZipDrop least profitable
Q6 Retention root cause	First-14-day experience → 30-day churn; lift tables + logistic regression	Early failures & summer start primary; weak packs contributing

2 Analytical Approach

Data & tooling

Eight related tables (swap events 3.88M rows; hourly station telemetry 1.49M; riders 20K; batteries 6.5K; tickets 44K; stations 152; city-day context; 12 fleet partners). All joins and aggregations run in **DuckDB SQL inside a Colab notebook** (Python, pandas, statsmodels, matplotlib), so the full 3.9M-row history is analysed with no sampling. Every number in this report, the dashboard and our LinkedIn post comes from one exported metrics file.

Metric definitions

- **Service-failure rate** = (no charged battery + abandoned queue + system error) ÷ all attempts. Rider cancellations are reported separately.
- **Contribution margin per completed swap** = revenue – energy (kWh × station tariff) – battery wear – station fixed cost (rent + maintenance ÷ that station's swaps that month).
- **Battery wear** = purchase cost × SoH lost per swap ÷ (initial SoH – 70% end-of-life), measured per pack inside the data window and applied by supplier and pack type.
- **New-rider early churn** = signed up on or after 1 Jan 2024 and last swap within 30 days of the first swap (first swap before 1 May 2025, so 60+ days are observable).

Methods

- Trend, segment and concentration (Pareto) analysis; telemetry joined to swaps at station-hour level.
- **Difference-in-differences** for the peak-pricing pilot: pilot vs control cities, before vs after, with station and month fixed effects and station-clustered errors.
- **Logistic regression** of new-rider churn on first-14-day experience, with plan, city and channel controls; χ^2 tests and dose-response checks.
- Text re-labelling of ticket comments (English + Hinglish keyword rules, typo-normalised).
- **Robustness notebook**: load-matched, lot-level, rider-panel and activity-adjusted re-tests of every claim (Appendix B).

Data cleaning: every brief warning tested

Issue	What we found	Decision
Test stations STN-TST-01/02	62,031 rows, ₹3.93M billed, 24h a day , not the "small, zero-value" set the brief describes	Excluded
Firmware v3.2.0 clock bug (10 Mar–14 Apr 2025)	139,490 events logged 5h30m early; billing confirms it (PEAK swaps in peak hours: 12% → 100%)	+5h30m shift
Offline-sync near-duplicates	Rapid repeats: offline 0.16% vs realtime 0.17%; 0 duplicates under three stricter definitions	No excess → none dropped
km outliers / SoC & SoH > 100%	15,526 odometer resets; 9,853 readings > 100%	NULL / clip to 100
City spellings	21 strings for 6 cities (Bombay, BLR, Gurgaon...)	Alias map
Telemetry gaps	3.1% hours partial/missing, mostly poor connectivity	NaN-aware, never 0
CSAT not missing at random	44% response if resolved <24h vs 22% if slower	Report by bucket
Ticket categories	9,460 'other' tickets are range complaints	Re-labelled
payment_mode	Single value for every row, incl. pay-as-you-go	Unusable; use plan_type
Tickets out of window · asset register	86 tickets outside the window; 1,438 'retired' packs still issued afterwards	Excluded · flagged

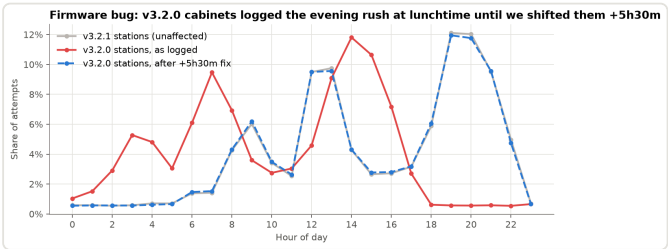


Figure 1. The v3.2.0 cabinets' demand curve lines up with unaffected stations only after the +5h30m correction.

3 Key Insights · Q1 Network performance over time

Volume nearly tripled, but failures spike every summer and the price-rise margin gain vanished within two months

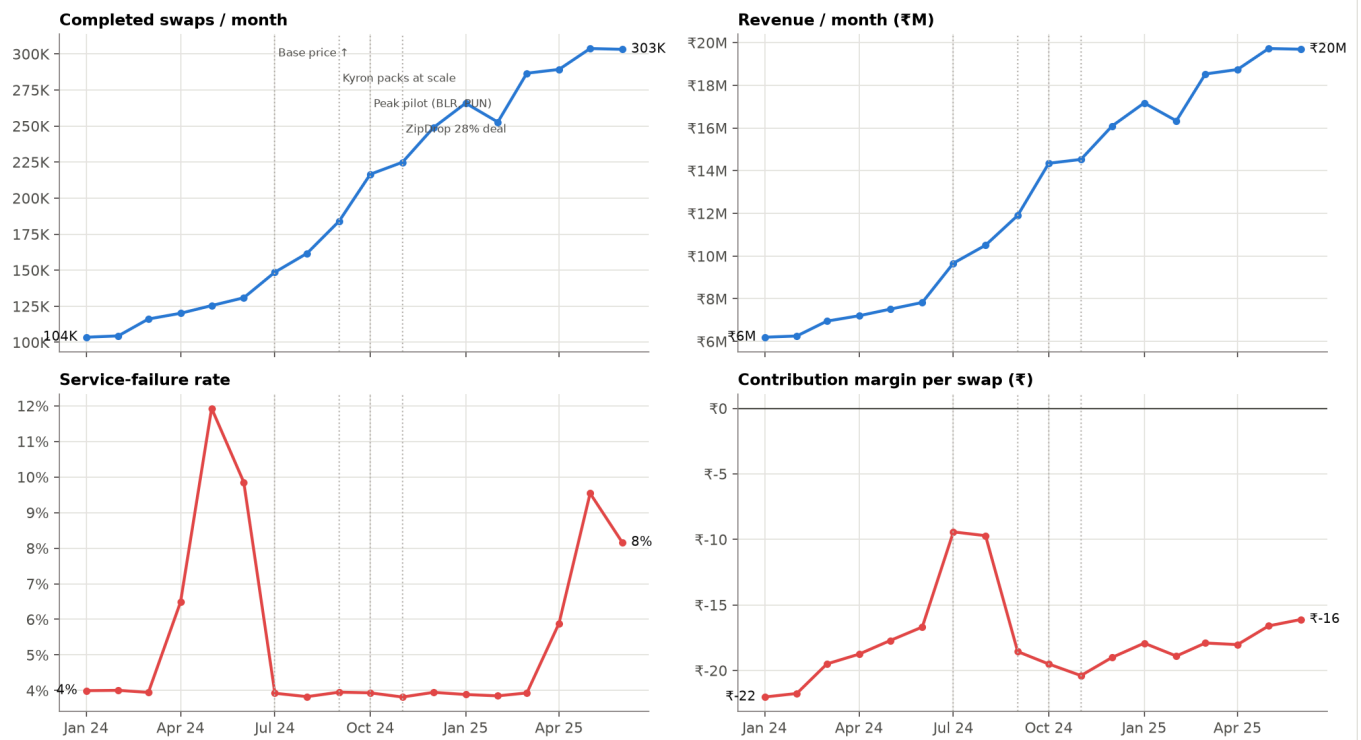


Figure 2. Monthly completed swaps, revenue, service-failure rate and fully-loaded CM per swap. Dotted lines mark the price rise, Kyrón at scale, the pricing pilot and the ZipDrop amendment.

The four metrics tell different stories. Volume grew steadily (104K → 303K completed swaps per month) and revenue kept pace (₹6.2M → ₹19.7M). Failures, however, sit at ~3.9% for nine months of the year and **spike every April–June** (11.9% in May 2024, 9.5% in May 2025), which is seasonal capacity stress, not gradual decay. Contrary to the brief, failures did not outgrow swaps: H1-2025 failure counts rose **2.1×** against **2.4×** for completed swaps. The summer problem has not worsened, but it has not been fixed either.

Margin: the July 2024 price rise lifted CM from –₹19.2 to –₹9.6 per swap, then CM **fell straight back to –₹18.2 from September 2024**, the month the bad Kyrón lots arrived at scale. Re-pricing those packs at a normal wear rate puts H1-2025 CM at –₹4.2 instead of –₹17.5: about **76%** of the loss is that one cohort.

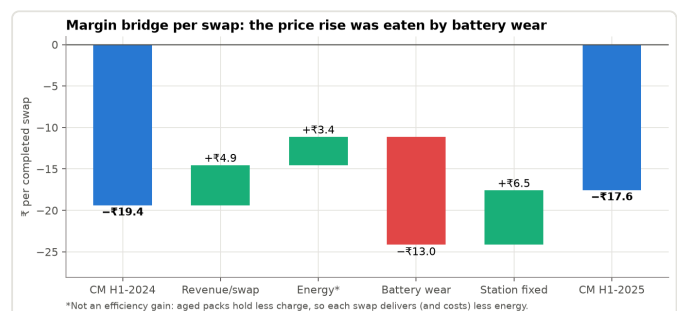


Figure 3. CM bridge, H1-2024 → H1-2025: +₹4.9 from price and +₹6.5 from scale, almost all absorbed by ₹13.0 of extra battery wear. The +₹3.4 energy step is not a gain: kWh per swap fell 2.02 → 1.69 because aged packs hold less charge.

SO WHAT Growth hides two separate problems: a seasonal *capacity* problem (failures) and an *asset* problem (margin, mostly one bad battery cohort). Neither is solved by more volume or more stations of the same kind. Sensitivity: the CM level depends on the wear assumption (with 30% salvage, CM went +₹2.3 → –₹2.5), but the direction and the driver hold under every assumption; before battery wear, margin per swap rose from ₹20 to ₹34.

3 Key Insights · Q2 Service failures & Q3 Stations

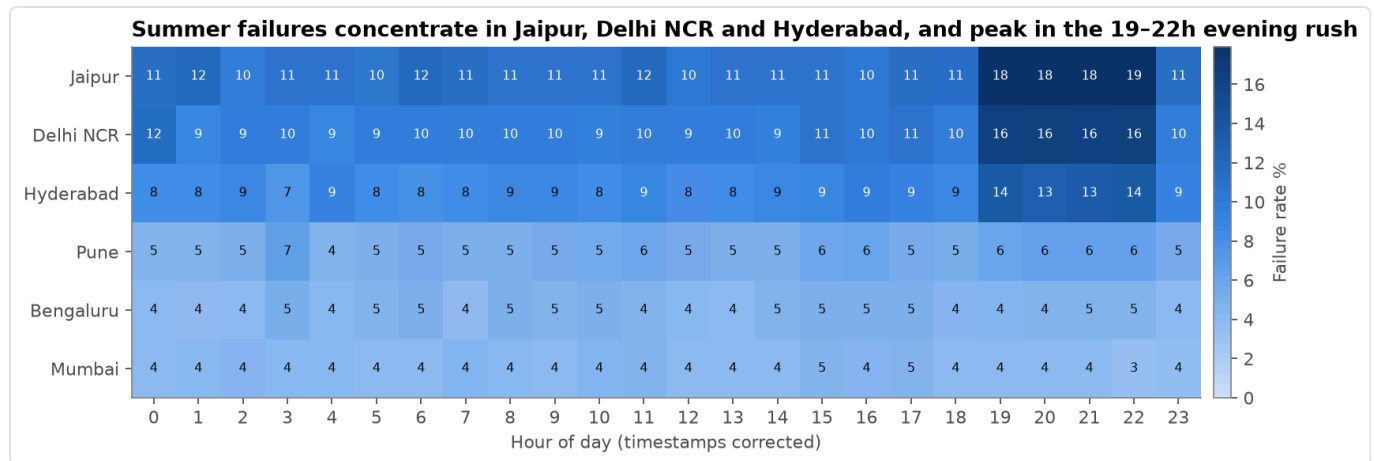


Figure 4. Summer failure rate by city and hour (timestamps corrected). Jaipur, Delhi NCR and Hyderabad fail 9–12% all day and 13–19% in the 19–22h rush; Bengaluru and Mumbai stay near 4%.

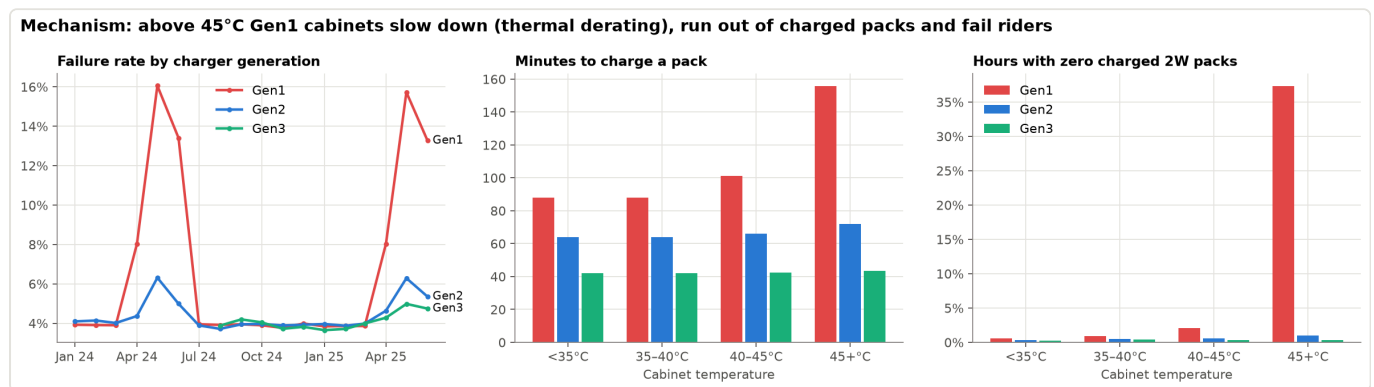


Figure 5. Mechanism from hourly telemetry: Gen1 charge time and stock-outs explode above a 45°C cabinet temperature; Gen3 is unaffected.

- **Supply, not software:** in summer "no charged battery" rises 2.4% → 5.5% and queue abandonment 1.3% → 2.7%, while system errors and cancellations are unchanged. When a cabinet's stock hits zero, failure doubles (11.2% vs 5.4%).
- **Concentrated:** the **36 hot-city Gen1 stations handle 32% of summer attempts but cause 57.5% of summer failures**; every one of the worst 20% of stations is Gen1.
- **3W:** 3W riders fail 12.2% in summer vs 7.9% for 2W, and the gap exists all year (8.6% vs 3.3%). It is not about 3W slots: failure is the same at stations with and without them.
- **Equipment, not just climate or load:** *within* Delhi NCR, Gen1 fails 16.5% of summer attempts vs 5.3% at Gen3 (Jaipur 16.8% vs 5.3%). At every load level Gen1 fails 14–16% vs ~5% for Gen3, and Gen3 sites are actually busier. In mild Bengaluru and Mumbai all generations perform the same.
- Location type, host type and connectivity move failure by ≤1 pt. Complaints follow the same map (Jaipur 14.5 vs Mumbai 8.9 tickets per 1,000 swaps).

SO WHAT Service quality is not spread evenly. It is a **Gen1-in-hot-cities** problem: Gen1 is 48% of attempts but 66% of summer failures, and all 51 Gen1 cabinets are still in service after two expansion waves. Failed riders don't retry, so every failure is a lost sale (≈₹14.3M over 18 months).

3 Key Insights · Q4 Batteries & equipment

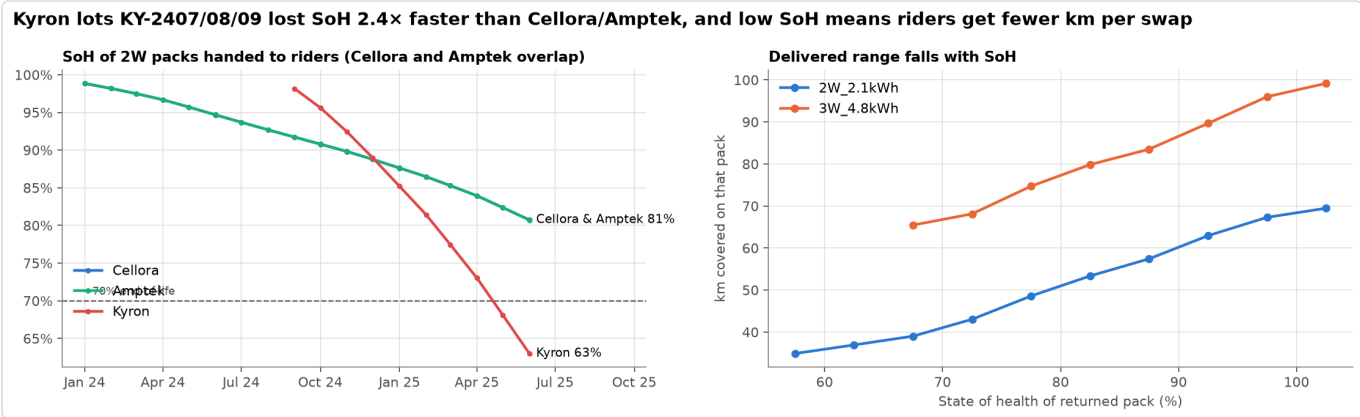


Figure 6. Left: SoH of 2W packs handed to riders. Kyron lots KY-2407/08/09 fall through the 70% end-of-life line within a year of entering service. Right: delivered range falls steadily with SoH.

Supplier · 2W pack	Packs	SoH lost / 1,000 swaps	Wear ₹ / swap	km per swap	Current SoH	Retired
Cellora	3,035	31.7	₹34.7	60.9	80.0%	0
Amptek	1,098	31.6	₹34.6	60.9	80.0%	0
Kyron (lots KY-2407/08/09)	1,385	75.2	₹82.3	48.6	60.5%	1,385 (all)

- A lot defect, not a supplier one:** the three bad lots lost SoH 2.4× faster, but Kyron's later 3W lots (KY-2410 to 2418) lose 33 points per 1,000 swaps, the same as Cellora. **BMS firmware makes no difference**, and the bad packs' SoH is identical in all six cities, so it is not heat.
- All 1,438 retired packs are Kyron, retired after 7–9 months (₹44.3M of 2W packs at purchase cost; the premature part is ≈₹28.7M). The retirement is **on paper only**: they were issued 33,761 more times, and 22% of June-2025 swaps used a pack below 70% SoH.
- Riders feel it: 2W range falls from ~67 km at 95% SoH to ~37 km at 60%, and at the *same* SoH a bad-lot pack delivers 11% fewer km. Battery/range tickets (incl. 9,460 mislabelled 'other' comments like "scooter weak lag raha hai") rose from 308 per quarter to **6,167**, and are now **37% of all tickets**.

SO WHAT One battery cohort is the **primary driver of margin erosion** (~76% of the H1-2025 loss) and a contributor to complaints and churn. "More batteries" only helps as *replacements* for the 1,385 bad 2W packs (25% of the 2W fleet), bought on lot acceptance tests rather than supplier name.

3 Key Insights · Q5 Pricing & partner economics

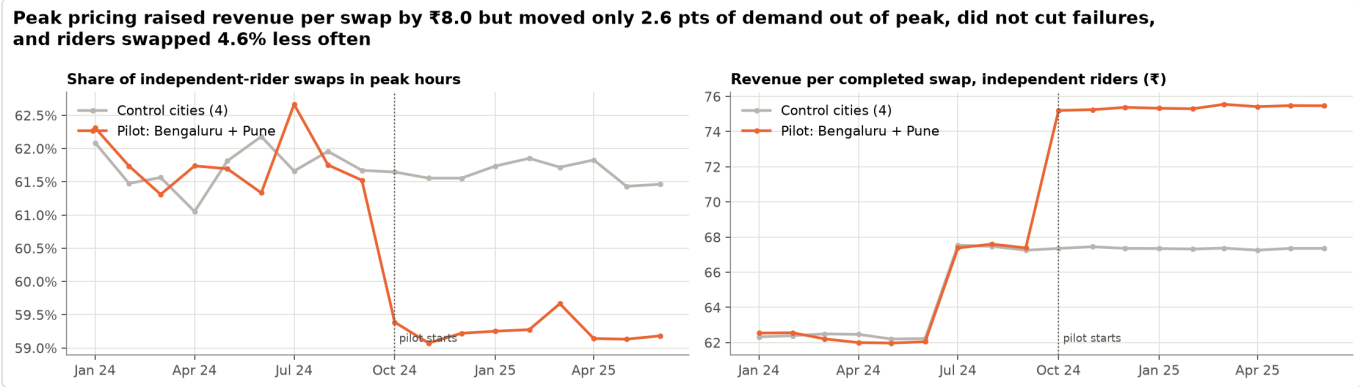


Figure 8. Independent riders, pilot cities (Bengaluru, Pune) vs four control cities. Revenue per swap jumps at pilot start; peak-hour share barely moves.

Intervention	Evidence	Effect	Who it hits
Base price ↑ (1 Jul 2024: 2W ₹60–65, 3W ₹100→110)	Before/after	+ ~₹5 revenue per swap; volume growth unchanged	Everyone (partners at their discount)
Peak/off-peak pilot (BLR, PUN, Oct 2024)	Difference-in-differences, station + month FE, clustered SE	+₹8.0 per swap (95% CI 7.7–8.3); peak share –2.6 pts ; failure rate +0.05 pts (no change) ; swaps per rider –4.6% , so revenue per rider +7.9%	Independents; surcharge-exempt partners (ZipDrop, Swiggle Go) untouched
ZipDrop amendment (Nov 2024: 12% → 28%, no peak surcharge)	Same partner before/after	Revenue/swap ₹57.5 → ₹48.6 ; CM/swap – ₹19.7 → –₹31.6	15% of network volume

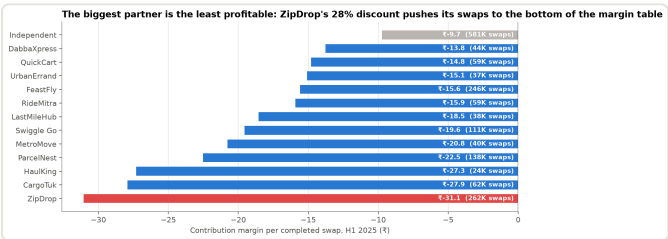


Figure 9. CM per swap by partner, H1 2025.

- Gig riders cannot move their shifts, so price is a **revenue lever, not a congestion lever**. The pilot left summer failures untouched, and some swaps simply left the network.
- The largest partner is the least valuable:** ZipDrop earns –₹31.1/swap vs –₹13.8 for the best partner (DabbaXpress) and –₹9.7 for independent riders. It loses –₹19.9 per swap even before site costs, and the deeper discount bought no volume (share 15.1% → 15.5%). The extra discount costs **≈₹5.7M a year**.
- 3W cargo partners are also margin-negative despite ₹110 pricing: 3W packs carry ₹77–81 of wear per swap.

SO WHAT Pricing and contracts affect segments very differently. Peak pricing earns more from independents; the ZipDrop deal gives away more than any other partner term.

3 Key Insights · Q6 Root cause of new-rider churn

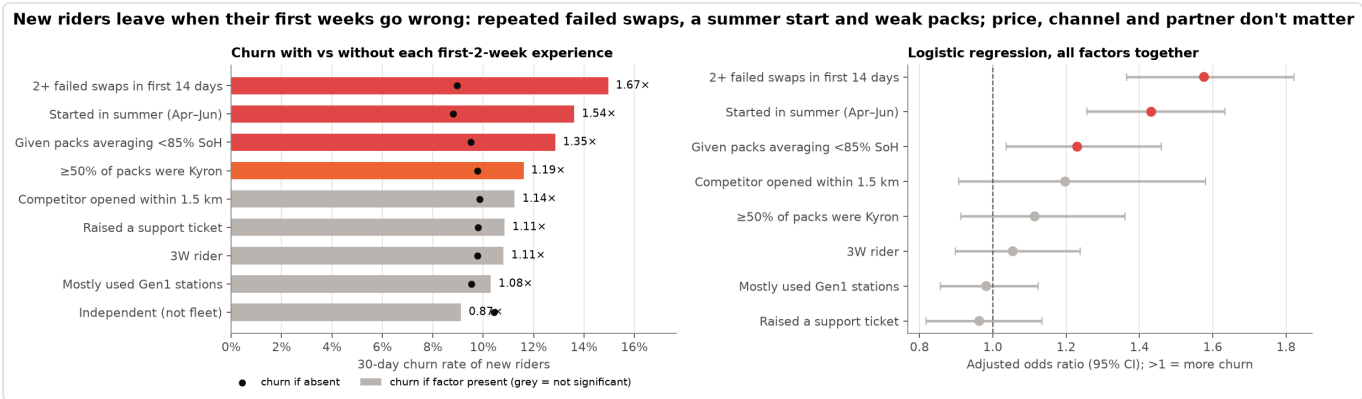


Figure 10. Left: 30-day churn with and without each first-2-week experience (13,797 new riders; base churn 9.9%). Right: adjusted odds ratios from a logistic regression with plan, city and channel controls.

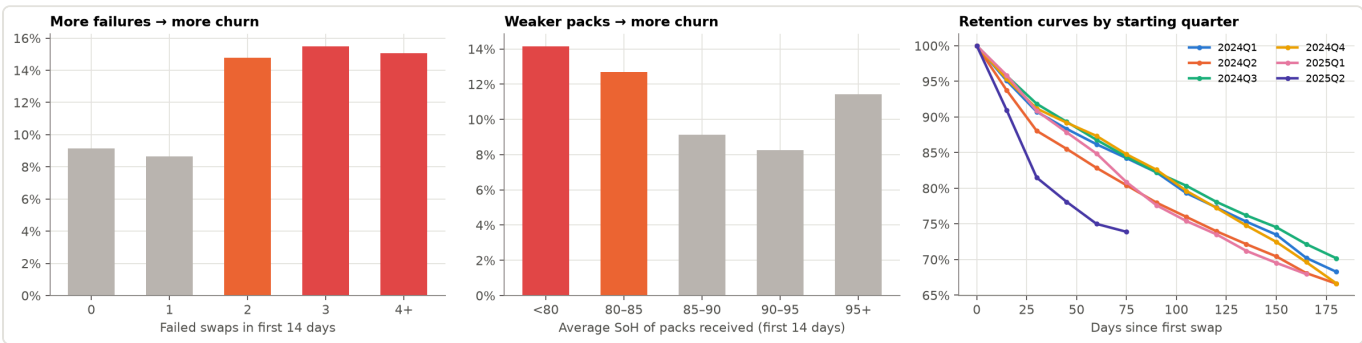


Figure 11. Dose-response and cohort curves: churn jumps once a rider suffers 2+ early failures; weaker packs raise churn; riders who start in Q2 (summer) leave fastest.

Tier	Factor	Evidence
PRIMARY	2+ failed swaps in the first 14 days	Churn 15.0% vs 9.0%; adjusted OR 1.58 (95% CI 1.36–1.82, $p<0.001$), 2.05 once early riding activity is controlled (heavy riders see more failures but churn less); riders whose early failure rate topped 20% churned at 23.3%; 16% of new riders exposed
PRIMARY	Starting in summer (Apr–Jun)	Churn 13.6% vs 8.8%; OR 1.43 even controlling for early failures; the April-2025 intake churned worst of all (16.8% vs 11.9% a year earlier)
CONTRIBUTING	Weak packs (avg SoH < 85%)	Churn 12.9% vs 9.5%; OR 1.23 ($p=0.02$), 1.46 with activity controlled; only 2025 riders ever received such packs
SECONDARY	Kyrón-heavy packs, nearby competitor, 3W, raising a ticket	1.1–1.2× raw lift; not significant once failures and pack health are controlled
NOT A DRIVER	Price plan, signup channel, fleet partner, KYC, Gen1 home station, competitor promotions	No significant effect (promotions: OR 0.90, $p=0.63$). Gen1 acts <i>through</i> failures (mediator), not on its own

SO WHAT Retention is decided by the first two weeks of *operations*, not by price or marketing. The same Gen1-in-summer and weak-battery problems behind Q2–Q4 are what push new riders out. (Churn is noisy, pseudo- R^2 0.015: these are the controllable levers, not the full story. The published odds ratios are lower bounds.)

4 Business Findings: primary vs secondary drivers

#	Finding	Role	Why we believe it
1	Gen1 chargers derate above 45°C → stock-outs → summer failures in Jaipur, Delhi NCR, Hyderabad	PRIMARY service failures	Mechanism visible in telemetry; holds <i>within</i> cities and at every load level; sharp threshold (4.1% → 18.7% failures as the day's cabinet peak passes 43–46°C); repeats in both summers
2	Kyron lots KY-2407/08/09 degraded 2.4× faster → lower range, 2.4× wear cost, ~76% of the H1-2025 margin loss	PRIMARY margin erosion · CONTRIB. churn, complaints	Lot-level split (later Kyron lots are normal); same SoH loss in every city; BMS ruled out; timing matches the Sep-2024 margin drop
3	Early failures and summer starts drive new-rider churn	PRIMARY churn	Logistic regression with controls, stronger when early activity is controlled; dose-response by failure rate
4	ZipDrop's 28% discount makes the largest partner the least profitable	SECONDARY network margin · key for partner strategy	Before/after on the same partner; negative even before site costs; the discount bought no volume
5	Peak pricing adds revenue but barely shifts demand and trims swaps per rider 4.6%	SECONDARY revenue lever only	Difference-in-differences with fixed effects; rider-month panel
6	Expansion waves added capacity evenly (+60–80% per city), not where failures were	CONTRIB. why summer 2025 ≈ 2024	Gen1 hotspot failure rates unchanged year on year (Delhi Gen1 16.4% → 16.6%)
7	Data issues: billed test stations, firmware clock bug, broken payment_mode, mislabelled tickets, 'retired' packs still in service	Governance	Direct checks; the ticket mislabelling hid the battery problem for months

The budget question, answered: the evidence does **not** support blanket growth. Of the four proposals, **batteries** (as replacements for the bad lots) and **stations** (as Gen1 → Gen3 swaps in three cities, cooling first) are supported; **peak pricing** only as a revenue tool after the capacity fix; the **ZipDrop exclusive** is not.

Proposal	Verdict	Evidence
More stations	Partly: only Gen1 → Gen3 replacement	New waves went everywhere; all 51 Gen1 cabinets remain and cause 66% of summer failures. Gen1 fails ~3× Gen3 at every load level, so more Gen1-style sites would not fix the heat problem.
More batteries	Yes, as replacements	Replace the 1,385 bad-lot 2W packs (25% of the 2W fleet, ≈₹44M): 49 km vs 61 km and ₹82 vs ₹35 wear per swap. Buy on lot acceptance tests, not supplier name; do not add packs for growth.
Network-wide peak pricing	Not as proposed	+₹8.0/swap, but riders swap 4.6% less, only –2.6 pts demand shift and no failure reduction; exempt partners dodge it.
ZipDrop long-term exclusive	No	15% of volume at –₹31/swap, the lowest margin on the network, and –₹19.9 even before site costs. An exclusive locks in the 28% discount.

Limitations. Observational (simulated) data: we show consistent associations backed by mechanism, timing and dose-response, not proof of causation. CM uses modelled battery wear and site costs, so its level is assumption-sensitive while its trend is not. Churn is defined on activity (no explicit cancellation). The pricing effect is estimated on independent riders only. Appendix B lists every claim we re-tested and what changed.

5 Actionable Recommendations

Action	Owner	Expected impact (sized from data)	KPI to track
1 Replace the 36 Gen1 cabinets in Jaipur, Delhi NCR and Hyderabad with Gen3 before April. Start with cooling retrofits on the same cabinets and pre-position charged packs before 12:00 on heat-alert days.	Network Ops	≈22,600 failures avoided / yr (39% of summer failures), ≈₹1.4M revenue; cooling by 5°C alone ≈14,400 (64% of that)	Summer failure rate at these sites (16% → ~5%); days with cabinet peak above 45°C
2 Pull and replace the bad-lot packs. Remove the 1,438 'retired' Kyrion packs still in use, replace the 1,385 2W packs before summer, claim warranty on KY-2407/08/09, and buy on lot acceptance tests, not supplier name.	Procurement / Asset	≈₹44M of replacement packs; ≈₹28.7M premature asset loss not repeated; +12 km per swap for affected riders	Wear ₹/swap by lot; share of swaps below 70% SoH; battery tickets
3 Protect every new rider's first fortnight: route new riders to healthy packs (screen by lot, since SoH readings flatter the bad lots), auto-credit after any failed swap, and hold back summer acquisition in hot cities until #1 is done.	CX / Growth	≈135–156 more riders kept per 13.8K new riders (≈1 pt churn), compounding with #1	30-day churn of new riders; % with 2+ early failures
4 Renegotiate ZipDrop; do not sign an exclusive on current terms. Move to a volume-tiered discount capped near 15%, make the peak surcharge billable, and tie discounts to off-peak volume.	Partnerships / Finance	Up to ≈₹5.7M / yr margin recovered	CM per ZipDrop swap vs network average
5 Keep peak pricing as a revenue tool. Extend it to the four control cities for independent riders after #1 lowers summer failures; deepen off-peak discounts if shifting demand is the aim.	Pricing	≈₹3.7M / yr extra revenue, net of 4.6% fewer swaps per rider	Revenue per rider; swaps per rider; churn of independents in rollout cities
6 Fix the data plumbing: exclude test cabinets, patch firmware v3.2.0, repair payment_mode, reconcile the asset register with cabinet logs, add a battery/range ticket category and a lot-level SoH alert.	Data / Eng.	Battery problems visible ~6 months earlier; trustworthy KPIs	Share of 'other' tickets; clock-drift alerts

90-day roadmap

- Weeks 0–4:** pull the 'retired' packs and order replacements on lot tests; warranty claim; new-rider pack routing and auto-credit; ticket taxonomy.
- Weeks 4–8:** Gen3 procurement and site survey for the 36 hot-city Gen1 cabinets; cooling retrofits as a stop-gap; ZipDrop renegotiation.
- Weeks 8–12:** first Gen3 swaps in Jaipur and Delhi NCR before the heat; summer inventory playbook; decide the pricing rollout on post-fix data.
- Governance:** one monthly dashboard (this one) owned by Ops, with summer failure rate at Gen1 sites, wear ₹/swap by supplier and new-rider 30-day churn as north-star KPIs.

A Appendix: supporting evidence

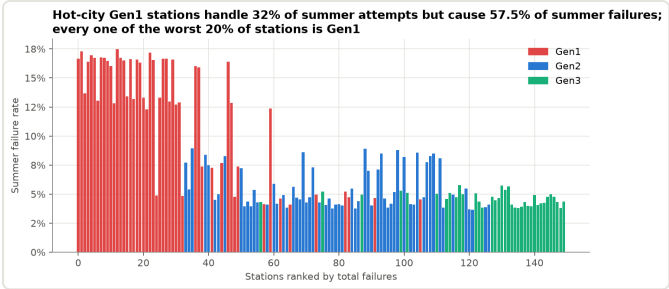


Figure A1. Stations ranked by failures; summer failure rate coloured by generation.

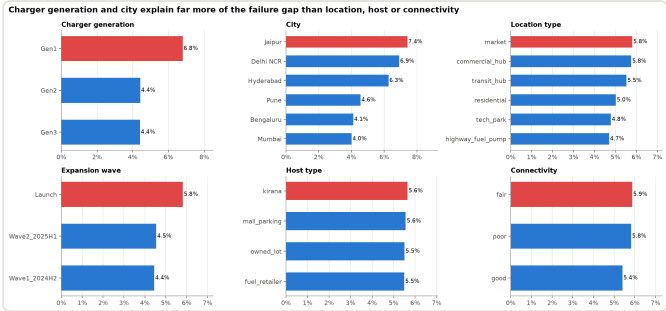


Figure A2. Failure rate by station attribute: generation and city dominate; location, host and connectivity matter little.

Unit economics, network average per completed swap (Jan 2024 – Jun 2025)

Revenue	Energy	Battery wear	Station fixed	Contribution margin
₹63.9	-₹19.5	-₹48.6	-₹13.4	-₹17.6

CM sensitivity to the battery-wear assumption

Period	Base (70% end-of-life)	With 30% salvage value	Before battery wear
H1 2024	-₹19.2	-₹7.4	₹20.3
Jul–Aug 2024 (after price rise)	-₹9.6	+₹2.3	₹29.8
Sep 2024 – Jun 2025 (bad lots at scale)	-₹18.2	-₹2.5	₹34.1

If the bad-lot packs had worn like a normal pack, H1-2025 CM would be -₹4.2 instead of -₹17.5 (base case).

Reproducibility

Notebook **VoltRelay_Analysis_WindRocker.ipynb** runs top to bottom in Google Colab (mounts Drive, reads the CSV or CSV.GZ files) or locally, and regenerates every figure, the dashboard data and outputs/metrics.json. The companion **VoltRelay_Robustness_Checks_WindRocker.ipynb** re-tests every headline claim and writes outputs/audit_metrics.json. Heavy aggregation in DuckDB; statistics in statsmodels/scipy. Synthetic dataset, generation seed 59500.

Team WindRocker · Gradient Learnings Data Analytics Hackathon · All company names are fictional.

B Appendix: how we tried to break our own findings

For every headline number we asked four questions: could something else produce the same pattern (**confounding**)? Is a "concentration" just who was observed longest (**exposure bias**)? Are two numbers really the same money (**double counting**)? Does a sizing assume nobody reacts (**behavioural response**)? The full tests are in the robustness notebook.

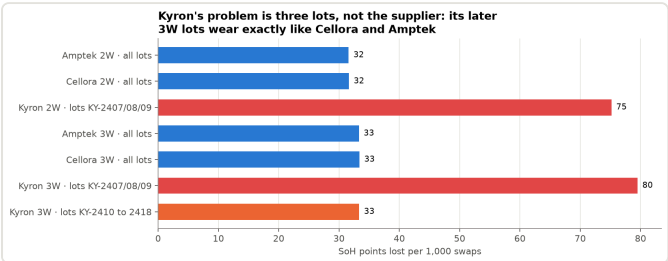


Figure B1. SoH lost per 1,000 swaps by supplier and lot. Kyron's later 3W lots wear like Cellora and Amptek: the defect is in three lots.

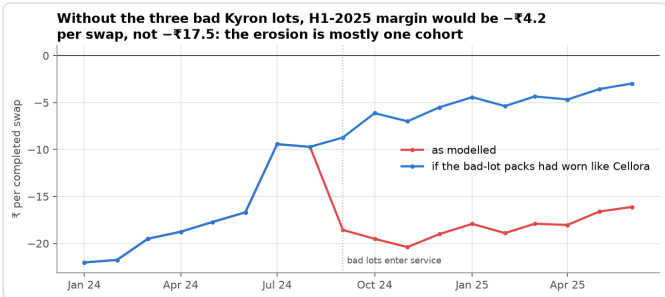


Figure B2. Monthly CM per swap as modelled vs re-priced at a normal pack's wear rate: most of the post-Sep-2024 erosion is the bad lots.

Claim	What we tested and found	Verdict
Gen1 × heat drives summer failures	Gen1 fails 14–16% in every load quartile vs ~5% for Gen3 (Gen3 is busier); outages, offline chargers and quarantined packs are negligible	Holds
"Worst 20% of stations = 43% of failures"	They carry 31% of attempts and were open all 18 months (1.4× concentration). Sharper: 36 hot-city Gen1 stations = 57.5% of summer failures	Reframed
"3W fails more because fewer stations have 3W slots"	3W failure is identical with or without listed 3W slots, and the gap exists all year; 57% of 3W swaps happen at "zero-slot" stations	Dropped
"Stop buying Kyron"	Only lots KY-2407/08/09 are bad; later Kyron lots wear normally (33 vs 33 SoH points per 1,000 swaps)	Now a lot problem
₹48M written off + ₹34M/yr excess wear	Same money counted twice. 2W packs: ₹44.3M gross, ≈₹28.7M premature; no recurring saving once the packs are pulled	Corrected
+₹3.4 energy saving in the margin bridge	kWh per swap fell 2.02 → 1.69 because packs aged (SoH 97% → 81%); riders got less energy	Relabelled
Pricing rollout +₹6.1M a year	Pilot riders swapped 4.6% less often (rider-month panel, rider + month fixed effects); revenue per rider +7.9%, not +11.8%	≈₹3.7M
Churn OR 1.58 for 2+ early failures	Heavy riders see more failures but churn less; controlling for activity gives 2.05 (weak packs 1.23 → 1.46)	Understated
Firmware fix, near-duplicates	Billing confirms the +5h30m shift (12% → 100% of PEAK swaps in peak hours); 0 duplicates under three stricter definitions	Holds
Competitor activity	Promotions: OR 0.90 (p=0.63); stations near a new competitor did not lose volume	Not a driver